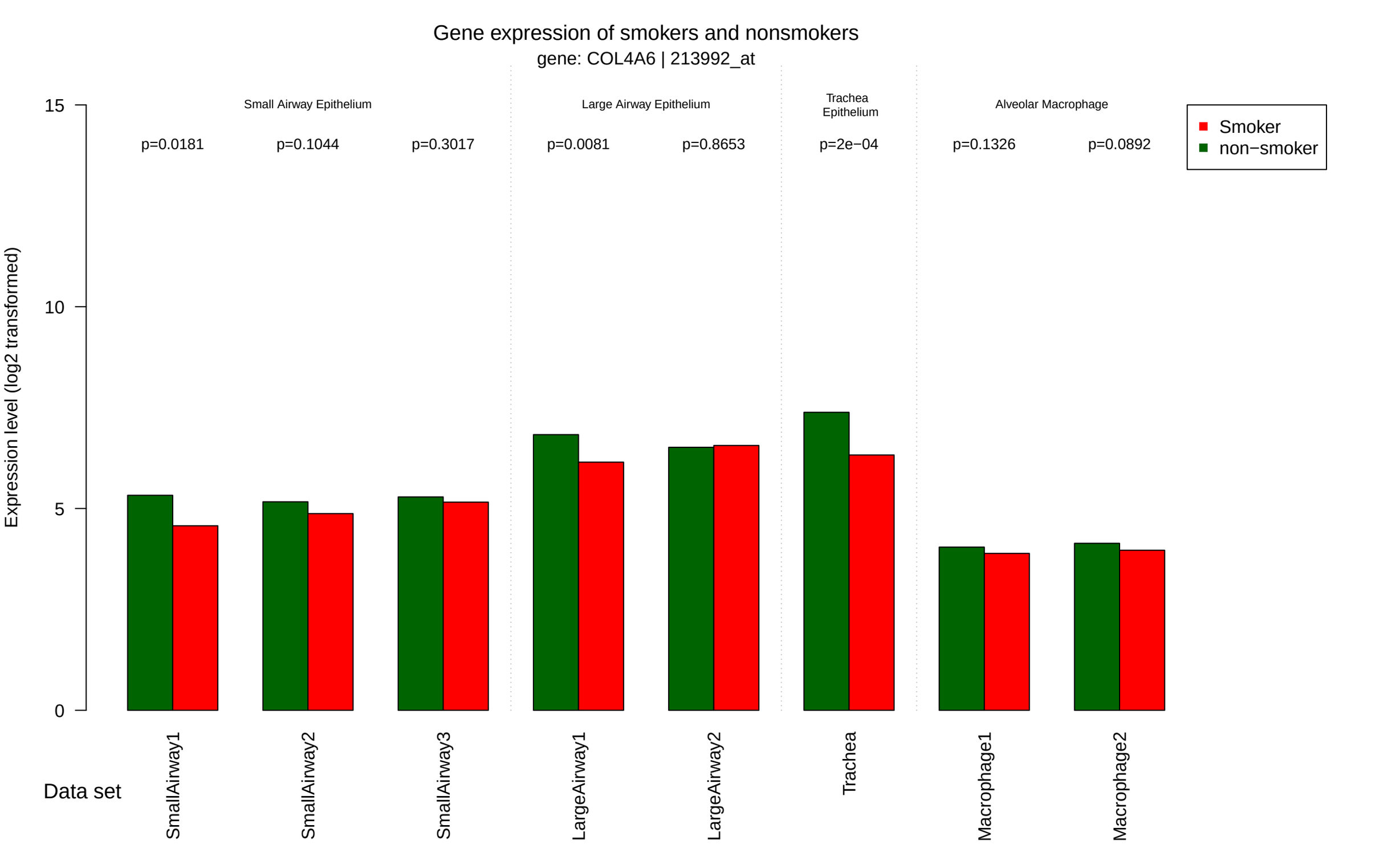
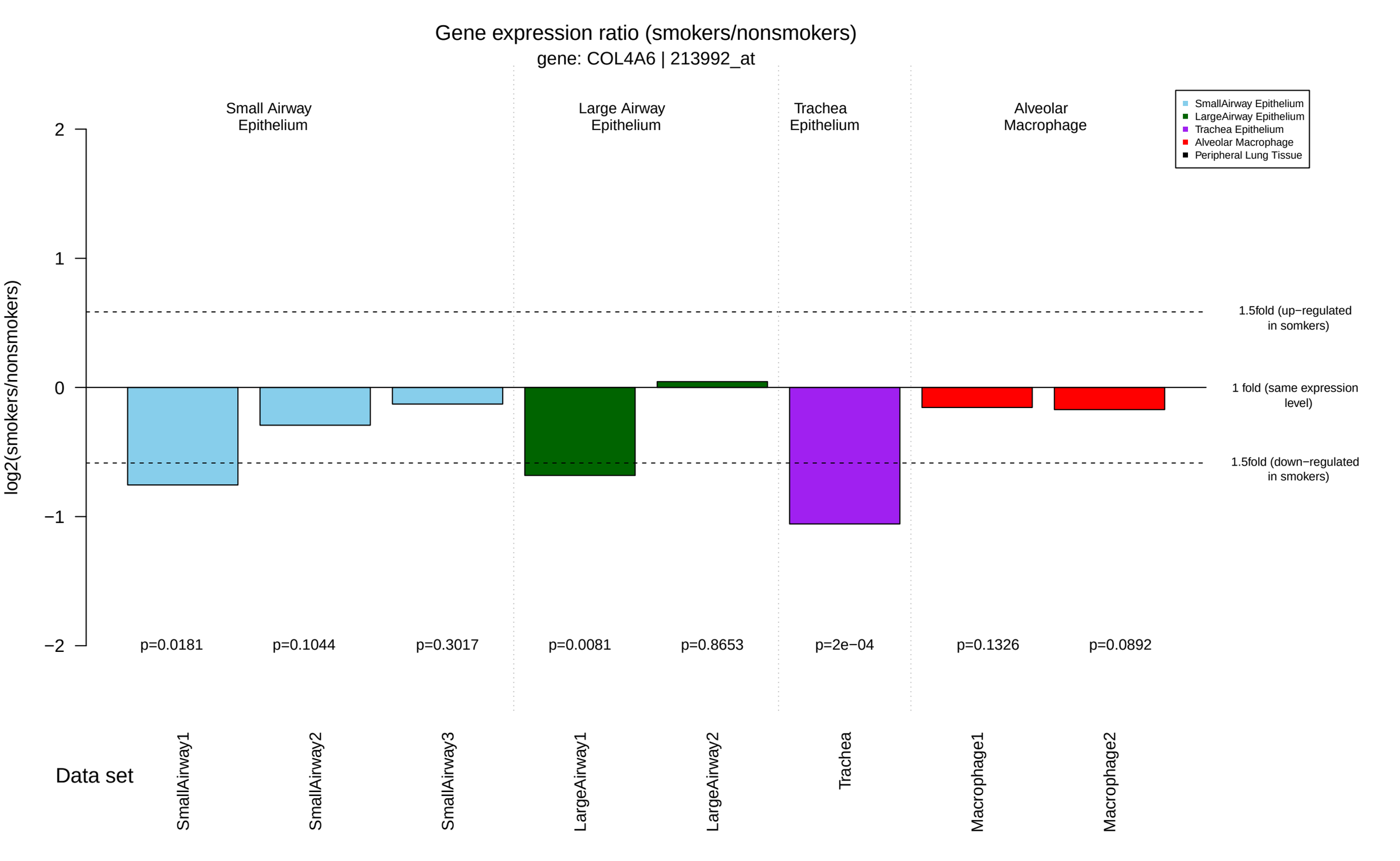
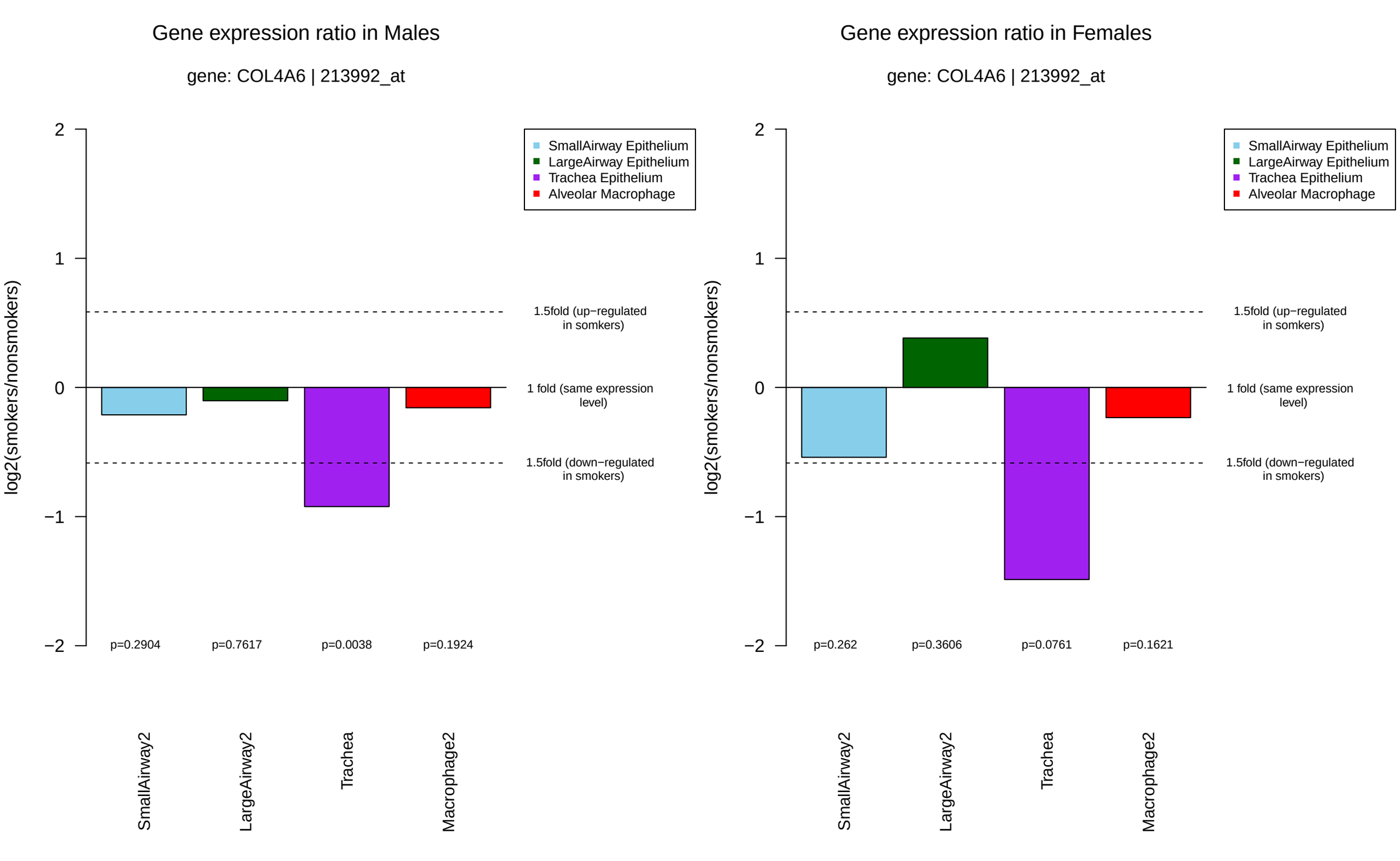
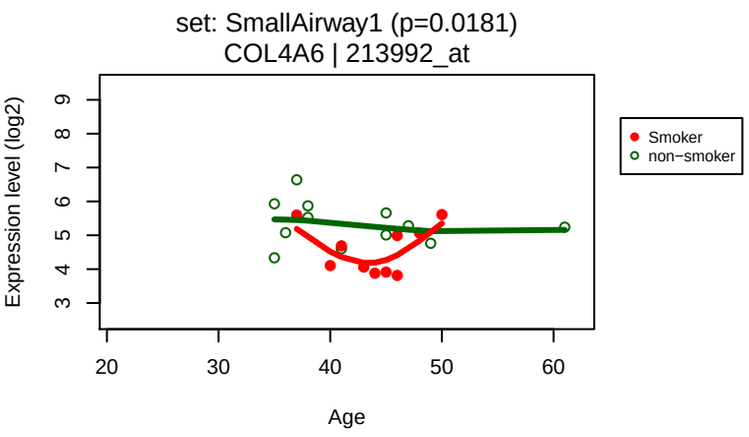
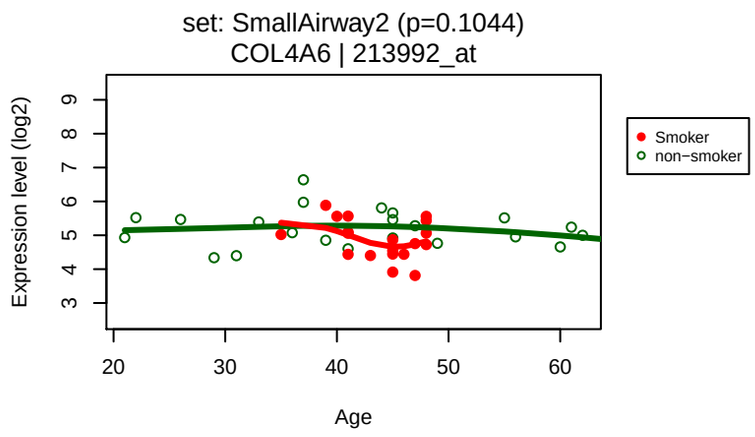
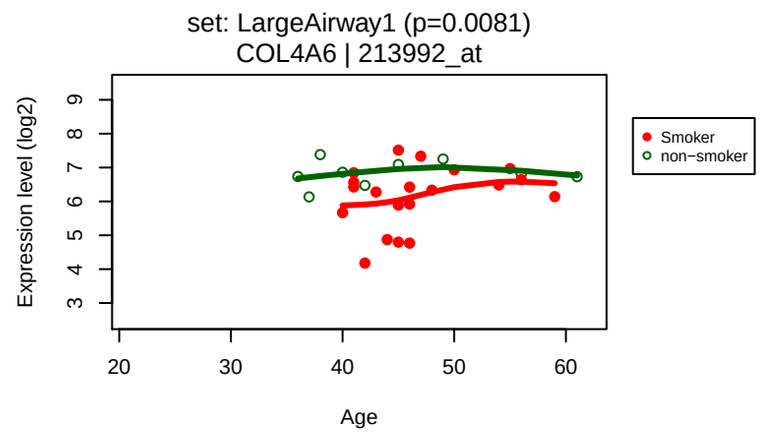

Gene expression of smokers and nonsmokers
gene: COL4A6 | 213992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0181
p=0.1044
p=0.3017
p=0.0081
p=0.8653
p=2e−04
p=0.1326
p=0.0892
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

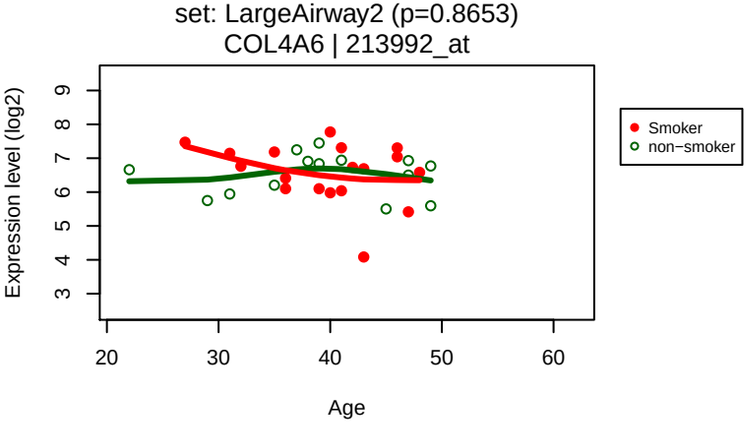

Gene expression ratio (smokers/nonsmokers)
gene: COL4A6 | 213992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0181
p=0.1044
p=0.3017
p=0.0081
p=0.8653
p=2e−04
p=0.1326
p=0.0892
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

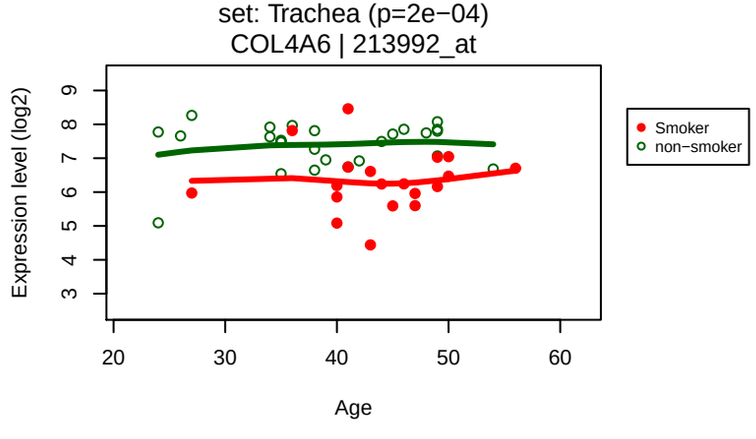

Gene expression ratio in Males
gene: COL4A6 | 213992_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2904
p=0.7617
p=0.0038
p=0.1924
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: COL4A6 | 213992_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.262
p=0.3606
p=0.0761
p=0.1621
SmallAirway2
LargeAirway2
Trachea
Macrophage2

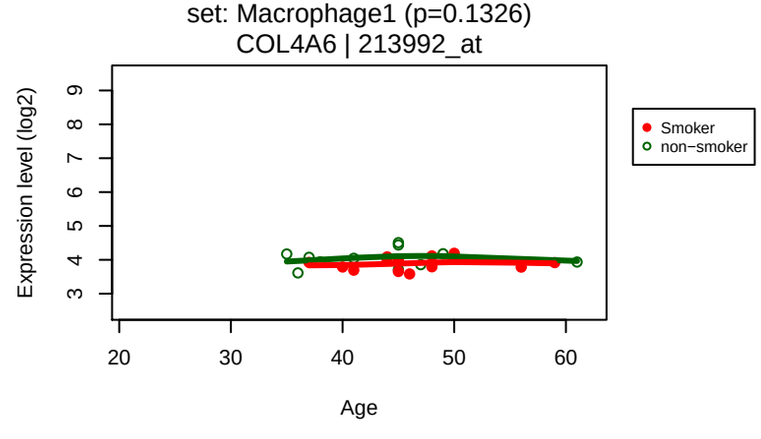

set: SmallAirway1 (p=0.0181)
COL4A6 | 213992_at
Expression level (log2)
Age
Smoker
non−smoker

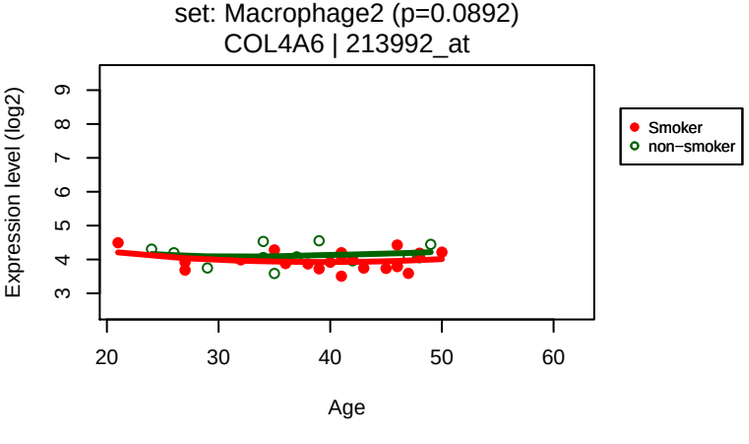

set: SmallAirway2 (p=0.1044)
COL4A6 | 213992_at
Expression level (log2)
Age
Smoker
non−smoker

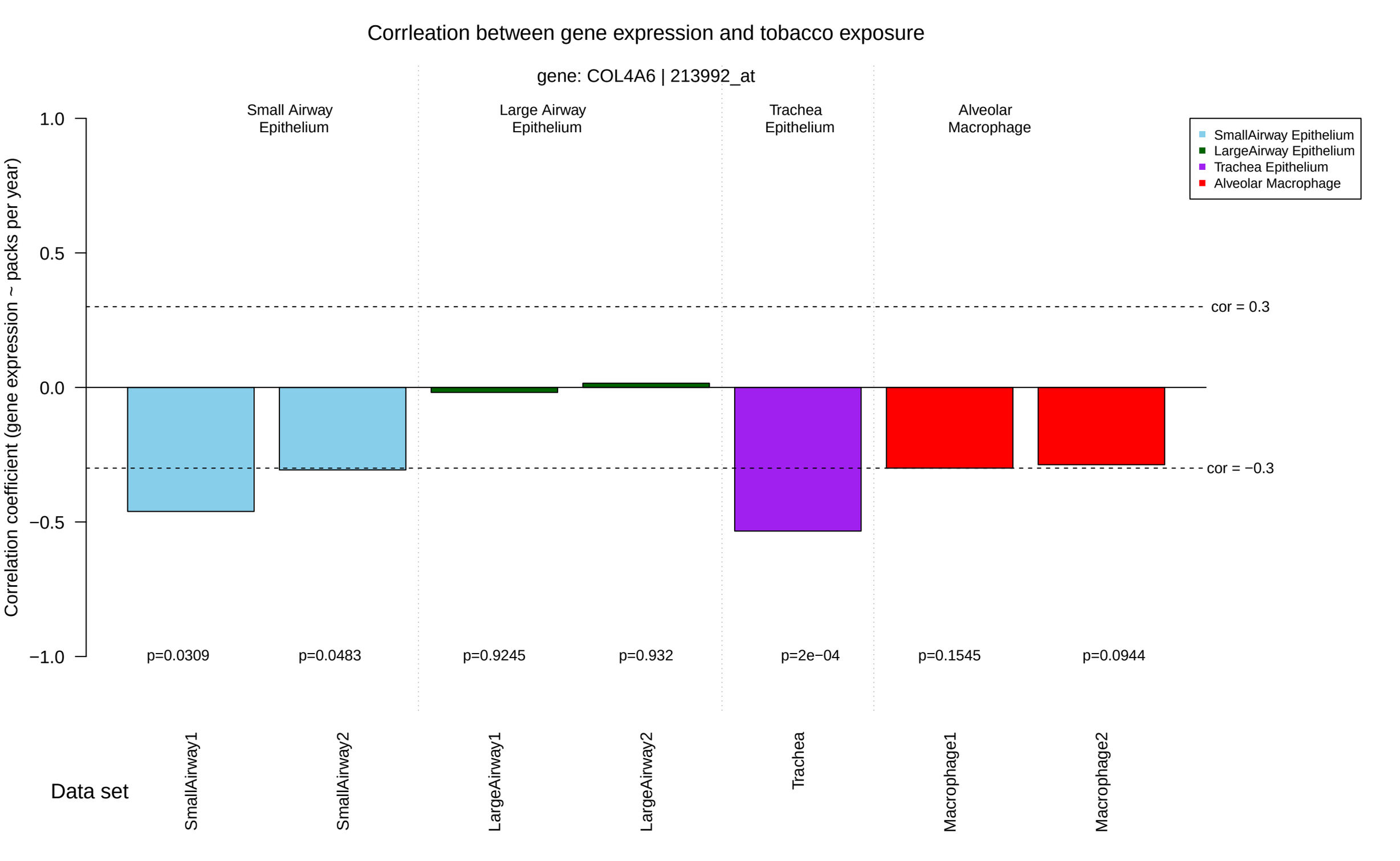

set: LargeAirway1 (p=0.0081)
COL4A6 | 213992_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.8653)
COL4A6 | 213992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=2e−04)
COL4A6 | 213992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.1326)
COL4A6 | 213992_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0892)
COL4A6 | 213992_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: COL4A6 | 213992_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0309
p=0.0483
p=0.9245
p=0.932
p=2e−04
p=0.1545
p=0.0944
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2